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ORIGINAL ARTICLES

Horticultural Characteristics of Chestnut Growing in Turkey

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Abstract

Chestnut is one of the important nut crops in Turkey. However, studies on production, marketing, and characteristics of chestnut growers is very limited in the literature. Also, the type of chestnuts production changes depending on the region. This is the result of being chestnut is a natural forest tree. To find out the growing characteristics of chestnut in Turkey a highly structured questionnaire study is done. The purposes of this study, therefore, are (i) to explore the typology of chestnut growers; (ii) to outline the characteristics of the farms, their production, care activities (such as weed management, fertilization, orchard floor etc., major insects and disease problems and their management), and harvest characteristics; and, (iii) to analyze marketing characteristics (storage, marketing channel, market structure). The bulk of the research data are collected from randomly selected 279 chestnut growers all around the Turkey. Descriptive statistics were used for summarize the research results. The results of per characteristic were presented with detailed according to the regions. For this aim, recommendations are also provided for improving chestnut production.

Keywords Fertilization · Growing care characteristics · Marketing · Weed management

Gartenbauliche Merkmale des Maronen-Anbaus in der Türkei

Schlüsselwörter Befruchtung · Wachstumseigenschaften · Marketing · Unkrautbekämpfung

Introduction

As reported by Uzun et al. (2015), major agroforestry practices in Turkey as well as in most part of the world are classified as silvoarable, silvopastoral, agrosilvopastoral and orchard intercropping areas. Production (63,750 tons per year) of chestnuts (*Castanea sativa* Mill.) in Turkey considered one of the centres of origin of chestnut (*Castanea sativa* Mill.) is very important for world markets (TUİK 2016). Chestnut is important for both horticulture and chestnut agroforestry in Turkey. Agroforestry systems include both traditional and modern land-use systems where trees are managed together with crops in agricultural settings. They are dynamic, ecologically based, natural resource management systems that diversify and sustain production in order to increase social, economic and environmental benefits for land users at all scales. Besides, agroforestry can contribute to improve the environment and the lives of people. One of the important plant for agroforestry is chestnut. Chestnut can be used for its nuts, timber and honey.

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The main chestnut growing areas in Turkey are the Black Sea, Marmara and Aegean regions and they represent 30.9 and 61%, respectively, of Turkey's chestnut production. Ecology, farm size, topography, species, cultivar and cultural practices are very important for productivity of chestnut. Strategies must be determined to main characteristics of chestnut production and researches must be performed to enhance chestnut production while decreasing production costs. However, studies on production and marketing characteristics of chestnut growers in Turkey are very limited in the literature. Our study was aimed to explore the typology of the chestnut growers, to outline production characteristics (common germplasm, farm characteristics, care activities such as weed management, fertilization, orchard floor management, etc., major insect problems and disease problems and their management, harvest characteristics) and to analyze the marketing characteristics (storage, marketing channel, processing price, market structure).

Materials and Methods

The bulk of the research data are collected from randomly selected chestnut growers by using a highly-structured questionnaire. Questionnaire studies were done in 12 provinces in four areas: the Marmara region, the Aegean region, the Middle-West Black Sea region and the East Black region. Regions and provinces were selected depending on the production quantity and total chestnut tree numbers. A total of 279 questionnaires were submitted in Turkey (Fig. 1).

The questionnaire was divided into five sections. These sections are typology of the chestnut growers and farm characteristics (age, education level of growers, slope of their orchard, ownership of chestnut orchard etc ...), growing care activities (location of stand/orchard, pruning, fertilizing, weed management, etc ...), harvesting, storing and marketing (storing type, type of marketing, type of harvesting, etc. ...), chestnut germplasm (what cultivars do they have, what cultivar(s) do they prefer etc ...), disease and pests (which pests and diseases are present in their stand and what are their frequency, etc ...). A totally of 49 questions were asked. Descriptive statistics were used to summarize the research results. When focusing on the relationship and differences, advanced parametric and non-parametric statistical techniques were used.

Results and Discussions

Typology of the Chestnut Growers and Farm Characteristics

We found that 66.7% of chestnut growers in Turkey were 45-years or older. Most were graduated from elementary school. Average size of the growers' families were 2 persons. Marmara region had the biggest chestnut stands. In this region, the management of forestry provided an easement for utilization of surrounding chestnut forests. In the East Black Sea region, the main crops are hazelnut and black tea plantations within the natural areas of the chestnut forest. Therefore, in this region farm/stand size for chestnut was small. In Turkey, chestnut orchards/stands are gener-

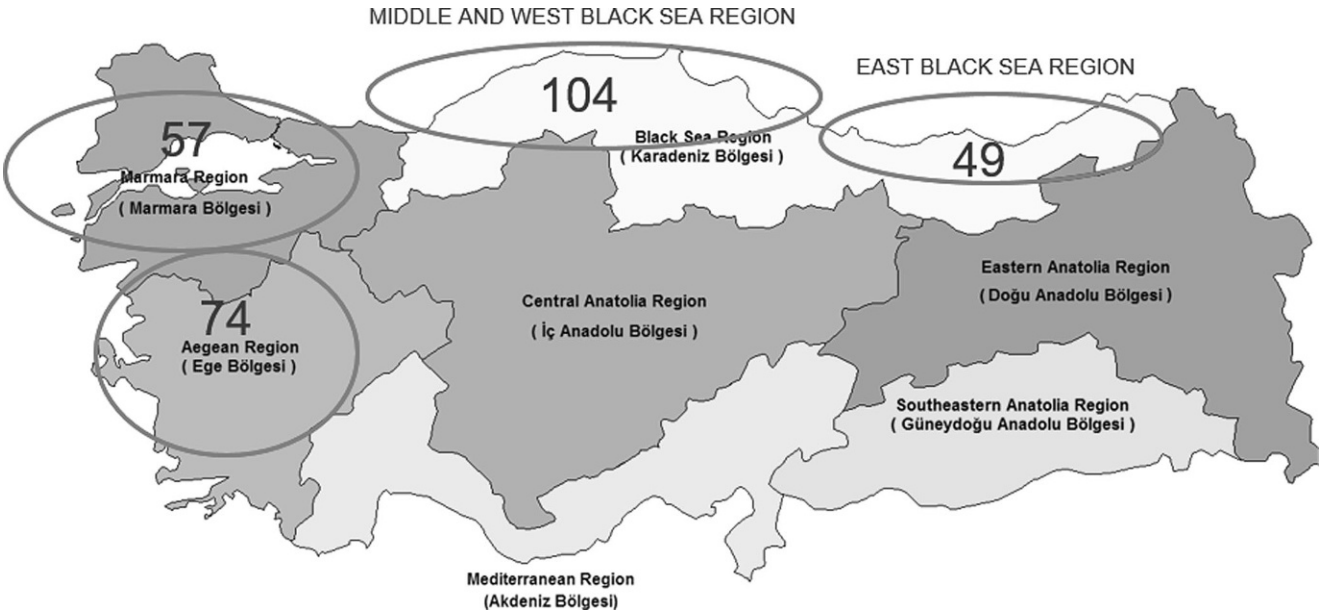


Fig. 1 Questionnaire numbers for regions

Table 1 Personal and farm characteristics of chestnut growers

Characteristics		Regions				Turkey
		Marmara	Aegean	Middle-West Black Sea	East Black Sea	
Age groups of growers (%)	18–31	7	1.4	7.8	20.4	8.2
	32–45	26.3	13.5	24.5	20.4	21.3
	46–59	38.6	40.5	37.3	34.7	37.9
	≥60	28.1	44.6	30.4	24.5	32.6
Education groups of growers (%)	Preschool	0	2.8	5.2	2.0	2.9
	Elementary school	87.7	87.5	90.7	53.0	82.7
	College	10.5	5.6	2.1	28.6	9.3
	Vocational school	0	2.8	1.0	6.1	2.2
	Bachelor degree	1.8	1.4	1.0	10.2	2.9
Family size (persons who stays continuously in the farm)		2	2	2	2	2
Farm size (da)		15.5	7.12	5.70	–	9.35
Parcel numbers		2.15	2.29	2.37	–	2.27
Orchards according to slope (%)	0–5	22.6	21.5	13.8	–	19.7
	6–20	49.2	30.7	49.1	–	44.6
	≥21	28.2	47.8	37.1	–	35.7
Growers get income from (%)	Fruits	100	100	100	89.7	97
	Wood	1.7	4	26.5	57.4	20
	Honey	21.0	1.3	39.7	44.8	26
Share of agriculture in total income (%)		73.7	72.6	69	46.1	67.7
Share of chestnut in plant production (%)		35.4	45.1	33.4	18.3	34.9
Farm purpose (%)	Grower	66.7	98.6	57.6	34.7	66.3
	Collector	–	–	11.1	53.1	13.3
	Ones both grower and collector	33.3	1.4	31.3	12.2	20.4
Ownership of chestnut orchards (%)	Private property	29.5	59.2	29.3	6.7	36.8
	Public forest	65.2	39.7	70.6	93.3	61.5
	Leasing	1.6	0.4	–	–	0.5
	Partnership	3.7	0.7	0.1	–	1.2
History of ownership (%)	Inheritance	64.0	54.5	92.8	100	68.9
	New established	26.7	29.6	4.4	0	21.9
	Bought	9.09	12.8	2.8	0	8.4
	Leased	0.21	3.01	0	0	0.8

ally located in areas with slopes. However, in the East Black Sea Region only 19.7% of chestnut orchards had a 0–5% slope (Table 1).

Share of agriculture in total income was about 67.7% in chestnut farms. Chestnut farms in the East Black Sea region had lower share of agriculture in total income than the other regions. Chestnut growers in the Aegean region get more income from chestnut in plant production. In Turkey, all chestnut growers had income from the nut production. However, some growers had also income from chestnut wood and honey. Although growers in the East Black Sea Region only represented 3% of chestnut production in Turkey, their total share of chestnut in plant production income was high (18.3%). Growers of this region are supported more from the production of chestnut honey and wood rather than nuts (Table 1). Share of growers in the Black Sea region who have income from chestnut honey was about two times that of growers in the Marmara region. This share was very limited (fewest) in the Aegean region.

In Turkey, there are two different methods to earn money from chestnut. First, 66.3% of the people involved with chestnut have income from chestnuts where they maintain and grow their trees. (Table 1). However, the second way is where some farmers had income by only harvesting nuts from chestnut forests (collector) (13.3%). These types of farmers only provide weed management before harvest. The biggest ratio of chestnut orchards/stands was found in the forest area, and they weren't private. Only 36.8% of chestnut orchards/stands were private property. The Aegean Region had the biggest ratio (59.2%) for chestnuts grown on

private property. Inheritance was responsible for the largest share for history of ownership. In Turkey, new orchards/stands were established generally in the chestnut forest areas 81.9% (Table 1). For this aim, growers generally grafted the seedling trees by top working. If the growers establish new chestnut orchards/stands in the field, they generally produce young plants on their own farm. Only 12.3% of the growers obtained chestnut plants from nurseries. Because of quarantine measures for chestnut diseases, grafted chestnut plant production was very limited in Turkey. The preference of some growers was not to produce chestnut seedlings on their own farm having purchased them from forest nurs-

Table 2 Production characteristics of chestnut growers

Characteristics		Regions				Turkey
		Marmara	Aegean	Middle-West Black Sea	East Black Sea	
Place of chestnut orchards (%)	Forest	96.5	54.1	92.0	100.0	81.9
	Field	1.8	36.5	0	–	11.8
	Both	1.8	9.5	8.0	–	6.3
Place of nursery plant obtained (%)	Own farm	92.9	82.6	100	–	87.7
	Nursery	7.1	17.4	0	–	12.3
Propagation method of bought nursery plants (%)	Grafted	66.7	9.1	0	–	21.4
	Seedling	33.3	90.9	100	–	78.6
Growers had grafted chestnut trees (%)		100	100	24.6	2.1	56.7
Growers who make pruning (%)		78.9	97.3	34.3	23.4	58.5
The type of pruners (%)	Regular pruners	10.8	23.1	2.9	10	14.5
	Only in some cases	89.2	76.9	97.1	90	85.5
Pruning tool (%)	Modern[a]	100	85.7	72.7	33.3	85.7
	Conventional[b]	0	24.4	27.3	66.7	27.8
Growers who applying manure (%)		17.5	32.4	6.1	0	14.4
Frequency of manuring (%)	Every year	0	55.0	33.3	–	37.1
	One in two years	88.8	40.0	50	–	54.3
	One in three years	11.2	5.0	16.6	–	8.6
Growers who fertilizing (%)		12.3	13.5	13.1	0	10.8
Frequency of fertilizing (%)	Every year	42.9	75.0	66.6	–	63.0
	Every other year	42.9	12.5	33.3	–	29.6
	Every three years	14.3	–	0	–	3.7
	Every five years	0	12.5	0	–	3.7
Growers who making orchard floor (%)		19.3	47.9	8.1	0	19.9
Frequency of making orchard floor	Every year	100	72.4	0	–	72.5
	One in two year	0	27.6	100	–	27.5
Growers who making weed management (%)		96.5	94.5	80.6	22.9	77.5
The type of weed management (%)	Conventional	61.1	66.7	48.6	45.4	57.5
	Motor scythe	38.9	33.3	51.4	54.6	42.5
Growers who making irrigate (%)		42.1	29.2	0	0	16.9
The type of irrigation (%)	Drip	26	33.3	–	–	28.2
	Spring	4.3	–	–	–	10.2
	Flooding	69.5	53.3	–	–	61.6
Frequencies of main diseases (%)	Chestnut blight	77.7	98.4	71.7	10.2	64.8
	Root rot	1.8	7.8	3	–	3.2
Growers make effort to protect trees against to chestnut blight (%)		64.9	24.3	27.3	–	29.4
Frequencies of main pests (%)	Chestnut weevil	68.5	37.5	49.5	6.1	40.5
	Grasshopper	11.1	0	0	–	2.1
	Aphid	1.8	0	0	–	0.3

[a]Pruner. Saw. Chainsaw.
[b]Girebi. Nacak. Tahra (Type of axes)

eries. Therefore, only 21.4% grafted chestnut plants were purchased from nurseries.

Growing Caring Activities

Pruning, fertilizing, manuring, irrigation, weed management and orchard floor management are important caring activities for chestnut growing. 58.5% of growers pruned chestnut trees in Turkey. Pruning activities were very high in the Aegean and Marmara regions, respectively 97.3 and 78.9%; however, growers generally pruned for dead and diseased branches. Growers who pruned their chestnut trees regularly represented only 14.5%. Generally, they use modern pruning tools (85.7%). But, in the East Black sea region they prefer conventional pruning tools like "girebi, nacak or tahra" (66.6%). Only, 14.4% of growers manured the chestnut trees in Turkey. For this activity, the Aegean region had the highest ratio (32.4%). (Table 2). Growers manured the chestnut trees when the trees were young. So, the share of growers who manured the chestnut trees regularly was 37.1%. 10.8% of growers fertilized the chestnut trees in Turkey. Growers in the East Black Sea region didn't fertilize the chestnut trees. Ratios for this characteristic were similar in the other regions. Although the fertilizing ratio was low in Turkey, the ratio of growers who fertilized regularly was high (63%) (Table 2). 19.9% of growers managed orchard floor in Turkey. Growers in the East Black Sea region didn't manage orchard floor in the chestnut orchards. Growers in the Aegean region had the highest ratio also for this activity. 77.5% of growers managed weeds in their chestnut orchards. Generally, chestnut growers prefer to manage weeds before harvesting. Ratios of growers undertaking weed management were 96.5, 94.5 and 80.6%, respectively in the Marmara, Aegean and Middle and West Black Sea region. Weed management was made with human power, animal power, motor scythe and through the use of herbicides in Turkey (Table 2). But the share of human power and use motor scythe were higher. 16.9% of growers provided irrigation in Turkey. There was no irrigation in the Black Sea region because of higher total rainfall each year. Growers irrigated generally by flooding. Ratio of drip irrigation was 28.2%.

Harvesting, Storing and Marketing

Chestnuts were harvested with poles (49.1%) or collected from the ground (50.4%) in Turkey. 48.4% of the growers did not store chestnuts; they generally collected chestnut from the ground and sold them immediately (Table 3). Most of the growers stored their chestnuts in open areas as burr stacks and covered with ferns or other plant materials (78.3%). These are called "Gömü" or "Yığın". In these places, growers stored their chestnuts and sold them depending to the market's demand at diffirent times. Only a few growers stored the chestnut in cold rooms. Growers directly sold their chestnuts to merchants or customers. Only a few growers sold them to cooperatives (4.5%) or chestnut candy companies (5.8%). Chestnuts were generally consumed after roasting. However, the chestnut processing sector has continued to develop and has progressed

Table 3 Harvest, store and marketing characteristics of chestnut growers

Characteristics		Regions				Turkey
		Marmara	Aegean	Middle-West Black Sea	East Black Sea	
Type of harvesting (%)	With pole	52.9	94.5	34.5	6.8	49.1
	Collecting from ground	46.8	5.5	65.5	93.2	50.4
	With machine	0.21	–	–	–	0.4
Growers who store chestnut (%)		22.2	83.3	54.7	18.4	51.6
The type of storing (%)	With burr	91.7	89.3	72.4	22.2	78.3
	With shell	8.3	10.4	21.3	55.5	18.0
	As kernel	–	0	5.8	22.2	3.9
The type of selling the chestnut (%)	Cash	90.6	91.7	100	100	95.7
	Dated	9.4	8.3	0	0	4.3
Marketing channel (%)	Consumer	10.7	8.4	49.5	85.7	30.4
	Merchant	73.2	95.7	62.6	7.1	71.6
	Cooperative	16	1.4	0	7.1	4.5
	Factory processing chestnut	19.6	4.2	0	–	5.8
Price ($)	Grafted	2.44	2.10	3.70	–	2.74
	Seedling	1.05	–	2.18	2.90	2.04

into important factors for chestnut markets in Turkey (Uylaşer et al. 2014). Also prices changed depending on the regions. Fresh chestnut prices per kg were higher in the Black Sea region (Table 3). Grafted cultivars had almost two times the value than seedling trees.

Chestnut Germplasm

All growers had grafted chestnut trees in the Marmara and Aegean regions (Table 2). However, this share was only 24.6% in the Middle and West Black Sea region and 2.1% in the East Black Sea region. Chestnut germplasm in the Black Sea region generally have the characteristics of easily peeling, seed coat does not enter the embryo and the chestnuts are extremely sweet. So, they are called as kuzu kestane (lamb chestnut). Fruit size of chestnut is not as important in this region as it is other regions, therefore, chestnut fruits are generally used for roasting. Chestnut germplasm of the Marmara and Aegean regions have generally larger fruit size, but peeling of the seed coat and taste are generally variable. In these regions, most cultivars are preferred especially for chestnut candy; however, some of them are preferred both for chestnut candy and roasting.

Many chestnut cultivars have been grown in the Marmara region. Eleven chestnut cultivars have been registered in this region in 1990 (TTSM 2016). Among these, 'Osmanoglu' and 'Sarıaslama' were the most frequently found cultivars in orchards. 'Osmanoglu' is an early ripening cultivar. It is preferred especially for processing into chestnut candy. Growers mentioned that this cultivar is less sensitive to disease although it is more sensitive to chestnut weevils. 'Sarıaslama' is less sensitive to chestnut weevils. It is preferred for both roasting and processing into chestnut candy. The cultivars 'Marigoule' and 'Maraval', hybrids of European and Japanese chestnut, have recently been introduced in the region. The cultivar 'Marigoule' has become a preferred choice. Growers mentioned that this cultivar is more tolerant to chestnut blight, drought and early ripening.

Some cultivars and genotypes are preferred in the Aegean region. Ozkarakas et al. (1995) selected 23 chestnut genotypes in the Aegean Region. Ertan et al. (2007) carried out a new selection study in the Nazilli district and five chestnut genotypes were selected for further studies. The most common cultivar in the region was 'Isıklar' (Sekerci). This cultivar was preferred for both roasting and processing into chestnut candy.

Seedling trees are generally used for chestnut production in the Black Sea region. Some selection studies have been carried out in the natural areas of European chestnut in the region (Ayfer et al. 1982; Serdar 1999, 2002; Serdar and Soylu 1999). An unregistered selection "Salipazari" (554-14 chestnut genotype) was a common local chestnut cultivar in Samsun province of the Middle Black Sea region. It was not officially registered because of its susceptibility to chestnut blight. However, five European chestnut cultivars were registered by Ondokuz Mayis University in 2009 (TTSM 2016). Among these, 'Erfelek' is grown in Sinop province of West Black Sea region. Cultivar 'Marigoule', has been tested since 2000 and registered for certification in young plant production (TTSM 2016). In this region, the cultivar 'Marigoule' was recommended for establishing chestnut orchards in the fields (Serdar et al. 2011) and new Anatolian cultivars 'Serdar', 'Erfelek' and 'Unal' were recommended for establishing chestnut stands in degraded forest areas. Also selection studies are still carrying out by Forestry Research Stations throughout the Anatolia.

Disease and Pests

In Turkey, chestnut blight was the most common disease (64.8%) on chestnut (Table 2). Growers made efforts to protect their trees against chestnut blight except the East Black Sea Region (Table 2). For this aim, in the Middle and West Black Sea Region, all growers made an effort cut the bark only of the infected branches. In the Aegean Region, some growers disinfect chestnut bark (24.3%) in addition to cutting the infected branches. In the Marmara Region, some growers disinfect chestnut bark and apply copper or lime, etc. (64.9%). On the other hand, specially, in the Marmara and Black Sea Regions, non-lethal infections appeared on chestnut trees and hypovirulent strains were found to be present (Akilli et al. 2009a, 2009b, 2011, 2012, 2013; Celiker and Onogur 2009; Erincik et al. 2008).

The frequency of root rot disease was only 3.2%, however it is becoming more pronounced. One reason the frequency of root rot was so low may be due to the fact that this disease may not be recognized as different from chestnut blight.

Chestnut weevil (*Curculio elephas* (Gyllenhal)) was the most important pest (40.5%) (Table 2). Growers generally tried to eliminate infested nuts with this pest. However, in the Marmara region, growers attempted to manage the Chestnut tortix (*Cydia splendana* (Hübner)) with pheromone or bait traps. Gall wasp (*Dryocosmus kuriphilus* Yasumatsu) was recently discovered in Turkey and now it is in a limited area (Ipekdal et al. 2014).

This study provided information on growing chestnut in Turkey genetic potentials of WBT populations in terms of nutritional properties. This information can be included recommendations for more productive chestnut farms in agroforestry systems. The nursery sector of the chestnut industry has not developed enough due to chestnut blight and root diseases in Turkey. Breeding studies should be performed in order to develop chestnut cultivars resistance to chestnut

blight. Seedling or clonal plants of European x Japanese hybrids should be used as rootstock in order to control root diseases. Establishing chestnut stands in suitable forest areas and fields should be encouraged. Biological control of chestnut blight should be used by growers when possible. Research will be performed to manage better chestnut weevil and nut rots. Grower cooperatives should be established in order to develop growing and marketing of chestnut. Storing of chestnut fruits in cold rooms should be encouraged. Also, further studies are required to determine the possibility of using chestnut trees as overstorey plant for new agroforestry systems such as agrosilvopastoral and home gardening or orchard intercropping in countries as Turkey.

Conflict of interest Ü. Serdar, B. Akyüz, V. Ceyhan, K. Hazneci, C. Mert, E. Er, E. Ertan, K.S.Ç. Savaş and V. Uylaşer declare that they have no competing interests.

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